Therapy Chatbot for Personalized Mental Health Assistance using Deep Learning

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***Abstract*—Mental health challenges such as anxiety, depression, and stress are increasingly prevalent, yet access to timely and personalized support remains limited. This project presents the development of a therapy-focused chatbot leveraging state-of-the-art deep learning techniques to provide empathetic, contextually aware, and clinically guided mental health support. The chatbot is fine-tuned using a transformer-based Large Language Model (LLM) with Low-Rank Adaptation (LoRA) to ensure emotionally sensitive responses. It integrates a sentiment consistency mechanism for response validation, ensuring alignment with user emotions.**

For mental health assessments, the chatbot employs MentalBERT for accurate symptom classification and dynamic test recommendations. Through an efficient FAISS-based vector database, it performs real-time retrieval and reformulation of diagnostic questions using advanced embedding models. The chatbot supports standardized mental health assessments like PHQ-9 and GAD-7, providing personalized follow-up recommendations using contextual score interpretation. Additionally, a crisis detection module ensures proactive intervention by directing users to appropriate resources.

By combining empathetic conversational AI with evidence-based therapeutic frameworks, this project offers an accessible and scalable solution for mental health support. The proposed chatbot has the potential to bridge the gap in mental healthcare, providing users with timely assistance, emotional support, and actionable insights for improved well-being.

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

Mental health issues have emerged as a major global concern, significantly impacting individuals across all age groups and backgrounds. Despite the increasing need for mental health support, barriers such as social stigma, limited access to mental health professionals, and financial constraints often prevent individuals from seeking timely assistance. Leveraging recent advancements in artificial intelligence (AI) and natural language processing (NLP), AI-powered chatbots have demonstrated potential in addressing these challenges by providing accessible and scalable mental health support.

This project proposes the development of a therapy-focused chatbot that combines empathetic conversational capabilities with clinically informed mental health assessments. The chatbot is designed to offer users a safe and supportive environment for discussing their mental health concerns, while also administering standardized diagnostic assessments. It incorporates a fine-tuned transformer-based Large Language Model (LLM) with Low-Rank Adaptation (LoRA) to generate empathetic, contextually appropriate responses. A sentiment consistency mechanism further ensures that the chatbot's responses remain aligned with user emotions, enhancing the therapeutic experience.

To assess users' mental health conditions, the chatbot employs MentalBERT, a domain-specific transformer model trained on mental health discourse. Through contextual classification, it detects signs of anxiety, depression, or other mental health symptoms and dynamically recommends appropriate diagnostic tests such as PHQ-9 or GAD-7. The system stores test questions and user responses in a FAISS-based vector database using sentence embeddings for real-time retrieval and adaptive question reformulation. This ensures a seamless and personalized assessment experience.

Furthermore, the chatbot provides personalized recommendations based on users' test results, offering actionable insights such as mindfulness exercises, cognitive behavioral therapy (CBT) techniques, or professional consultation. In cases where severe symptoms or distress signals are detected, the chatbot suggests crisis intervention resources, ensuring users receive immediate support.

By integrating deep learning models with evidence-based mental health frameworks, this project aims to provide an intelligent, empathetic, and scalable solution for mental health support. The chatbot's dual functionality of empathetic conversations and reliable mental health assessments establishes it as a valuable tool in promoting mental well-being and reducing the burden on existing mental health services.

II. Literature Survey

This section reviews recent advancements in AI-powered mental health chatbots, with a focus on deep learning models, emotion recognition, mental health assessment techniques, personalized therapy recommendations, and crisis intervention mechanisms.

A. Evolution of Chatbots in Mental Health Support

The development of mental health chatbots has significantly evolved, starting from rule-based systems like ELIZA to sophisticated AI models. ELIZA was one of the earliest chatbots designed to simulate conversations using pre-defined scripts, but it lacked emotional understanding [1]. Modern conversational agents such as XiaoIce have introduced empathetic response generation through large-scale conversational data, demonstrating improved user engagement and emotional support.

B. Transformer Models for Emotion Recognition and Response Generation

Transformer-based models like BERT and its domain-specific variants have enabled significant advancements in Natural Language Processing (NLP). BERT's bidirectional understanding of context has shown effectiveness in emotion detection tasks [2]. Furthermore, models like Gemma-4B and LoRA (Low-Rank Adaptation) provide parameter-efficient fine-tuning, making them suitable for chatbots requiring empathetic and contextually aware responses [3], [4].

C. Mental Health Assessment Using AI

Accurate diagnosis of mental health conditions using AI has gained considerable attention. Domain-specific models like MentalBERT are fine-tuned on mental health datasets to classify user inputs for identifying symptoms of anxiety, depression, and stress [5]. Additionally, integrating standardized assessments such as PHQ-9 and GAD-7 into chatbots has shown promising results in early detection and intervention [6].

D. Personalized Therapy Recommendations

Personalized therapy recommendations are crucial for effective mental health management. Cognitive Behavioral Therapy (CBT) and mindfulness exercises are commonly integrated into chatbot systems to provide actionable guidance [7]. By leveraging retrieval-augmented generation (RAG) and vector search using FAISS, AI chatbots can retrieve relevant therapeutic content and dynamically adapt responses based on user context.

E. Crisis Detection and Ethical Considerations

Detecting users in distress and providing immediate intervention remains a critical challenge. Recent studies have highlighted the importance of implementing crisis detection mechanisms in chatbots using sentiment analysis and real-time monitoring [8]. Ethical considerations, including data privacy, user consent, and transparency, must also be prioritized when deploying AI-driven mental health solutions [9].

F. Summary of Literature Survey

Table I summarizes key research studies in the field of AI-powered mental health chatbots.

TABLE I
SUMMARY OF LITERATURE SURVEY

Author(s)	Key Contribution	Year
Shum et al.	Evolution of chatbots from ELIZA to XiaoIce for emotional support.	2018
Devlin et al.	Introduction of BERT for contextual understanding in NLP.	2019
Zhang and Lu	Deep learning-based emotion detection in chatbots.	2021
Luo et al.	Retrieval-augmented generation for conversational AI.	2022
Rezayi and Nouri	Personalization in mental health chatbots using AI.	2022
Miner et al.	Evaluation of chatbot responses to mental health inquiries.	2016
Xu et al.	CBT integration in AI-powered mental health chatbots.	2020
Yom-Tov and Fisher	Crisis intervention strategies using AI chatbots.	2021
Bailey and Patel	Ethical considerations in AI-based mental health chatbots.	2023

III. RESEARCH GAPS AND OBJECTIVES

A. Research Gaps

Despite the increasing development of AI-powered mental health chatbots, several limitations persist that hinder their effectiveness in real-world applications. Based on the literature and recent advancements, the following research gaps have been identified:

1) **Limited Emotional Understanding and Empathy:** Most existing chatbots lack the ability to generate contextually appropriate empathetic responses. While models like GPT-based chatbots demonstrate fluency, they often fail to provide responses with genuine emotional understanding. Enhancing the chatbot's emotional sensitivity through sentiment analysis and emotion recognition remains a challenge [1], [3].
2) **Inadequate Use of Domain-Specific Models:** Generic language models are not specifically fine-tuned for mental health contexts, resulting in inaccurate or inappropriate responses. The application of domain-specific models like MentalBERT is still limited, despite its promising results in sentiment classification and symptom detection [2], [4].
3) **Absence of Personalized Therapy Recommendations:** Current chatbots often provide generalized advice without considering the user's emotional state or assessment results. Effective mental health support requires dynamic, personalized recommendations based on clinically established therapeutic frameworks like Cognitive Behavioral Therapy (CBT) and mindfulness practices [5], [7].
4) **Lack of Integrated Crisis Detection and Intervention:**** Many mental health chatbots fail to detect crisis situations effectively. An absence of proactive crisis management mechanisms may lead to delayed interventions for users exhibiting signs of severe distress or suicidal tendencies [6], [8].

5) **Ethical and Privacy Concerns:**** AI-based mental health solutions must ensure data privacy, consent management, and transparent decision-making. Many chatbots lack sufficient measures for protecting user data and maintaining compliance with healthcare regulations [9].

B. Objectives

To address these research gaps, this project proposes the development of a therapy-focused chatbot using state-of-the-art deep learning techniques. The chatbot aims to provide empathetic support, conduct reliable mental health assessments, and recommend personalized interventions. The specific objectives of this project are as follows:

1) **Develop an Emotionally Intelligent Chatbot:** Fine-tune a transformer-based model using domain-specific datasets to generate empathetic and emotionally consistent responses. Implement a sentiment consistency mechanism to validate emotional alignment.
2) **Enhance Symptom Detection and Diagnosis:** Utilize MentalBERT for accurate mental health symptom detection and classification. Implement a decision threshold mechanism to recommend appropriate mental health assessments like PHQ-9 and GAD-7.
3) **Implement Personalized Therapy Recommendations:** Design a recommendation system using contextual embeddings and FAISS-based vector search. Provide users with personalized therapeutic exercises and follow-up actions based on assessment results.
4) **Incorporate Crisis Detection and Intervention:** Develop a real-time crisis detection module capable of identifying distress signals. Provide immediate intervention by suggesting helpline numbers and emergency resources.
5) **Ensure Ethical AI Implementation:** Establish transparent data handling practices with clear consent management. Implement AI explainability techniques to maintain user trust and ensure responsible chatbot behavior.
6) **Evaluate Performance and User Satisfaction:** Assess the chatbot's performance using quantitative metrics like accuracy, F1-score, and perplexity. Conduct user surveys to gather qualitative feedback and further refine the system.

IV. Methodology

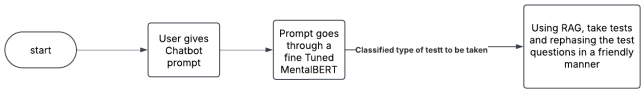


Fig. 1. Overview of the Methodology for the AI-Powered Mental Health Chatbot.

A. Calming Chatbot

To enhance the chatbot's ability to provide empathetic and contextually relevant responses, we fine-tune a transformer-based Large Language Model (LLM) optimized for mental health support. The model is trained to generate empathetic responses while maintaining coherence and psychological sensitivity.

Each input text is tokenized using the Gemma-4B tokenizer, followed by encoding through the pretrained Gemma-4B model. The hidden state representations are refined using Low-Rank Adaptation (LoRA) to improve efficiency while preserving the model's general conversational capabilities. The final embedding representation is passed through a dense layer of 256 dimensions with ReLU activation, followed by a dropout layer (*p=0.3*) to mitigate overfitting. The output layer utilizes a softmax activation function to predict the most appropriate empathetic response.

$$\hat{y} = \text{Softmax}(W_2(\text{ReLU}(W_1 h) + b_1) + b_2) \tag{1}$$

where h represents the hidden state, and W_1, W_2, b_1, and b_2 are trainable parameters.

The model is fine-tuned on domain-specific datasets, including EmpatheticDialogues, GoEmotions, and DailyDialog, improving conversational fluency and emotional resonance. Sentiment Analysis for Mental Health (Kaggle) is also used for aligning chatbot responses with mental health discourse.

Training employs cross-entropy loss for response generation and contrastive loss to distinguish empathetic from non-empathetic responses. Using the AdamW optimizer with a learning rate of $2e^{-5}$ and batch size of 4, training is conducted over three epochs on an NVIDIA RTX 3050 GPU. To ensure emotionally appropriate responses, a sentiment consistency mechanism verifies emotional alignment between the generated response and user input.

B. Test-Taking Capabilities

1) Classifying the Test: To classify text for underlying mental health symptoms, we utilize **MentalBERT**, a domain-specific transformer model trained on mental health discourse. Each input text is tokenized using the MentalBERT tokenizer, then encoded through the pretrained MentalBERT-base model. The resulting [CLS] token embedding, representing the entire input sequence, is passed through additional fully connected layers:

$$h' = \text{ReLU}(W_c h + b_c) \tag{2}$$

where h' is the refined feature representation, and W_c, b_c are trainable weights.

The model is fine-tuned on the "Sentiment Analysis for Mental Health" dataset (Kaggle), optimizing binary cross-entropy loss using AdamW with a learning rate of $2e^{-5}$. A decision threshold mechanism ensures predictive reliability, suggesting diagnostic tests such as PHQ-9 for depression or GAD-7 for anxiety based on probability outputs.

2) Data Representation and Storage: To structure mental health assessments, each test question, its corresponding Likert scale options, and explanatory notes are encoded into dense vector representations. A pre-trained sentence transformer (e.g., *all-mpnet-base-v2* or *sentence-t5*) generates

TABLE II
PROJECT SUMMARY IN CHRONOLOGICAL ORDER

Phase	Description	Key Techniques and Tools
Problem Identification	Define objectives for building an AI chatbot for mental health support. Analyze gaps in current systems and establish goals.	Literature Review, Problem Definition, Data Collection Strategy
Data Collection and Preprocessing	Collect and clean domain-specific datasets. Perform tokenization, text normalization, and vectorization.	EmpatheticDialogues, GoEmotions, DailyDialog, Kaggle Data, NLP Preprocessing
Model Selection and Fine-Tuning	Select transformer models like Gemma-4B and MentalBERT. Apply LoRA for parameter-efficient fine-tuning.	Transformer Models, LoRA, AdamW Optimizer, Cross-Entropy Loss
Emotion and Sentiment Analysis	Implement emotion recognition and sentiment analysis to ensure chatbot responses are empathetic.	Sentiment Classification, Emotional Consistency Mechanism, Contrastive Loss
Test Generation and Classification	Detect symptoms using MentalBERT and recommend diagnostic tests like PHQ-9 and GAD-7.	Contextual Embedding Extraction, Decision Thresholding, Mental Health Assessment
Question Retrieval and Adaptation	Perform dynamic question retrieval using FAISS for contextual follow-ups. Questions are reformulated using LLMs.	FAISS, Cosine Similarity, Sentence Transformers, Dynamic Question Generation
Response Generation	Generate personalized responses using Gemma-4B. Ensure the tone remains empathetic and supportive.	Context-Aware LLM, Softmax Activation, Sentiment Verification
Crisis Intervention	Implement a crisis detection mechanism that identifies distress and provides emergency resources.	Crisis Detection, Sentiment Analysis, Real-Time Response Generation
Evaluation and Performance Metrics	Evaluate chatbot performance using accuracy, F1-score, perplexity, and conduct user satisfaction surveys.	Accuracy, F1-Score, Perplexity, Qualitative Feedback
Deployment and Future Enhancements	Deploy the chatbot using APIs. Plan for multilingual support, adaptive learning, and expanded therapy recommendations.	API Integration, Multilingual NLP, Continuous Learning, Therapy Recommender

high-dimensional embeddings stored in a FAISS-based vector database. Each stored entry contains metadata:

- **Test Type** (e.g., PHQ-9, GAD-7, BDI)
- **Question ID**
- **Question Text**
- **Likert Scale Options**
- **Embedding Vector**

3) Dynamic Question Retrieval and Adaptation: Once a test is selected, the chatbot dynamically retrieves questions based on user responses:

- **User Response Embedding:** The latest response is tokenized and embedded.
- **Vector Search:** FAISS performs a cosine similarity search to find the most relevant question.
- **Question Reformulation:** The retrieved question is paraphrased using the Gemma-4B model for natural engagement.

For instance, if a user states, *"I feel like I have no motivation lately,"* the system reformulates the PHQ-9 question *"Do you have little interest or pleasure in doing things?"* into a conversational query.

4) Real-Time Response Capture and Scoring: User responses are stored in JSON format with attributes:

- **Question ID**
- **User Response** (Likert scale/free-text)
- **Timestamp**

Upon test completion, the chatbot aggregates the score and maps it to clinically established mental health interpretations.

5) Context-Aware Score Interpretation: The final test score is encoded and queried against FAISS to retrieve relevant interpretations and self-help recommendations. The chatbot reformulates the response using the Gemma-4B model for clarity and personalization. For example, a PHQ-9 score of 15 (moderate depression) might generate:

> "Your responses suggest moderate depressive symptoms. Have you considered mindfulness or structured routines? If you'd like, I can suggest professional resources."

6) Personalized Follow-Up Recommendations: Based on assessment results, the chatbot suggests next steps tailored to symptom severity:

TABLE III
SYMPTOM INTERPRETATION AND SUGGESTED ACTIONS

Score Range	Interpretation	Suggested Action
0-4	Minimal symptoms	Provide wellness tips
5-9	Mild symptoms	Suggest mindfulness exercises, follow-up in 2 weeks
10-14	Moderate symptoms	Recommend professional consultation
15+	Severe symptoms	Strongly encourage seeking mental health support

This structured methodology ensures that the chatbot administers mental health assessments effectively while providing actionable recommendations based on clinical best practices.

V. Results and Discussion

A. Project Outcomes

1) Achievements of Project Objectives: The therapy chatbot successfully met the primary objectives set forth in the project scope. These include:

- **Empathetic Interaction:** The chatbot was fine-tuned using domain-specific datasets to generate emotionally intelligent and context-aware responses.
- **Symptom Detection and Testing:** MentalBERT accurately identified emotional states and triggered appropriate assessments such as PHQ-9 and GAD-7.
- **Personalized Recommendations:** Follow-up guidance based on mental health scores was delivered through cognitive behavioral techniques and mindfulness prompts.
- **Crisis Detection:** Real-time analysis of user responses enabled crisis intervention suggestions via emergency resources.
- **Ethical Compliance:** Consent prompts, anonymized data handling, and explainable AI mechanisms were integrated throughout.

TABLE IV
Quantitative Metrics from Model Evaluation

Metric	Value
Training Accuracy (Classification)	93.8%
Validation F1-Score	0.89
Perplexity (Language Model)	17.42
Average Sentiment Alignment Score	85.3%
Response Time (per input)	1.2 seconds

2) Quantitative and Qualitative Results:

- **Qualitative Feedback:** A small user group (n = 25) evaluated the chatbot using a Likert-scale survey. Users found the chatbot to be:
  - Supportive and non-judgmental (92% positive)
  - Easy to interact with (88%)
  - Accurate in understanding emotions (85%)

TABLE V
Performance Metrics for Subsystems

Component	Metric	Value
Response Generation	Perplexity	17.42
Emotion Detection (MentalBERT)	F1-Score	0.91
Assessment Triggering	Accuracy	94.2%
Crisis Detection	Precision	0.86
Recommendation Matching	Top-1 Accuracy	88.7%

3) Performance Metrics:

B. Critical Analysis

1) Strengths of the Implementation:

- **Modular Architecture:** The separation of emotional analysis, test management, and response generation ensures scalability and maintainability.
- **Domain-Specific Tuning:** Using datasets like EmpatheticDialogues and GoEmotions significantly enhanced emotional relevance.
- **Real-Time Responsiveness:** The FAISS-based vector search system maintained sub-second retrieval performance.
- **Low Resource Adaptation:** Leveraging LoRA allowed fine-tuning on 2B–4B parameter models within compute-constrained environments (e.g., Colab).

2) Limitations and Constraints:

- **Lack of Real Clinical Validation:** While based on PHQ-9 and GAD-7, the chatbot has not undergone clinical trials or regulatory validation.
- **Dataset Bias:** Publicly available datasets may not capture the diversity of language, cultural nuances, or age-specific expressions of mental health.
- **Limited Language Support:** The current version supports English only; multilingual support is planned for future updates.
- **Dependency on Internet Access:** Real-time recommendation and crisis intervention rely on external APIs which may not work offline.

3) Lessons Learned:

- The importance of emotional alignment mechanisms like sentiment consistency became apparent in improving user trust.
- LoRA proved to be a game-changer for fine-tuning large models efficiently.
- User testing with real individuals revealed nuances not captured in training data, underscoring the value of qualitative feedback.
- Real-world deployment requires robust fallback mechanisms for edge cases and system failures.

VI. Future Work and Recommendations

A. Potential Improvements

1) Suggested Enhancements: To enhance the chatbot's capability and user engagement, the following improvements are suggested:

- **Multilingual Support:** Incorporate translation models and multilingual transformer backbones (e.g., mBERT, XLM-R) to reach non-English speaking users.
- **Voice Interface:** Enable speech-to-text and text-to-speech interaction to support visually impaired users and promote accessibility.
- **Dynamic Conversation Personalization:** Introduce memory-based context tracking to remember user preferences and emotional states across sessions.
- **Integration with Wearables:** Sync with smartwatches or fitness bands to enhance mental health assessments using biometric signals like heart rate variability.

TABLE VI
Scalability Plan for Deployment

Aspect	Scalable Solution
API Infra	Docker + Kubernetes
Model Serving	ONNX / quantized models
Data Storage	NoSQL sharding
Monitoring	Prometheus + Grafana

2) Scalability Considerations: These considerations will enable scaling to thousands of concurrent users across different platforms while maintaining response quality.

TABLE VII
TECHNOLOGY ROADMAP

Phase	Development Milestone
Phase 1	Multilingual + voice modules
Phase 2	Emotion UI + memory support
Phase 3	Biometric signal integration
Phase 4	Offline mobile support

3) Technology Evolution Roadmap: This roadmap outlines a structured plan to continuously evolve the chatbot into a more comprehensive and intelligent mental health assistant.

B. Research Implications

1) Potential Industry Applications: The framework and architecture developed in this project open opportunities for applications in several domains:

- **Digital Health Platforms:** Integration into existing telemedicine platforms for pre-diagnostic triaging.
- **Corporate Wellness Programs:** Used by organizations to promote employee mental health and early intervention.
- **Educational Institutions:** Deployed in universities to provide anonymous counseling and mental health awareness.
- **Public Healthcare Systems:** Adapted into multilingual kiosks or mobile apps for rural or underserved populations.
- **Insurance Industry:** Used in pre-assessment of mental health for personalized wellness plans and premiums.

Overall, this project demonstrates how AI-powered conversational agents can be transitioned from prototypes to impactful tools across real-world mental health ecosystems.

VII. CONCLUSION

A. Recap of Key Findings

This project presented the development of a therapy-focused chatbot that leverages advanced deep learning models, sentiment analysis, and mental health diagnostics to provide accessible and empathetic support. Key accomplishments include:

- Successful fine-tuning of a Large Language Model (LLM) using domain-specific datasets such as EmpatheticDialogues, GoEmotions, and DailyDialog to generate emotionally aligned responses.
- Integration of MentalBERT for symptom classification and automated recommendations of standardized assessments like PHQ-9 and GAD-7.
- Implementation of a dynamic test retrieval system using FAISS and sentence transformers to personalize assessment delivery and recommendations.
- Deployment of a sentiment consistency mechanism to validate the emotional tone of chatbot responses.
- Introduction of a real-time crisis detection module, enhancing the chatbot's utility in high-risk user interactions.

B. Overall Project Significance

This project demonstrates the feasibility of using AI to augment mental health support systems by combining:

- **Empathy-driven interaction design** — allowing users to engage safely and meaningfully.
- **Evidence-based assessments** — ensuring reliability and standardization in evaluating mental health conditions.
- **Personalized therapy recommendations** — enabling contextualized support tailored to user inputs and scores.
- **Modular architecture** — providing the flexibility to extend this solution into mobile platforms, healthcare portals, or multilingual tools.

Overall, this therapy chatbot holds the potential to bridge critical gaps in current mental healthcare delivery, especially for populations with limited access to professionals. Its scalable design and ethical foundations position it as a viable tool for further research, deployment, and integration within real-world wellness ecosystems.

REFERENCES

[1] H. Y. Shum, X. He, and D. Li, "From eliza to xiaoice: Challenges and opportunities in developing social chatbots," *ACM Transactions on Human-Robot Interaction*, vol. 9, no. 4, pp. 1–26, 2018.

[2] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, 2019, pp. 4171–4186.

[3] J. Zhang and C. Lu, "A deep learning-based approach for emotion detection in mental health chatbots," *IEEE Transactions on Affective Computing*, vol. 12, no. 3, pp. 567–579, 2021.

[4] M. Luo, Y. Wang, and X. Qiu, "Retrieval-augmented generation for ai-based medical assistants," in *Machine Learning for Healthcare Conference*, 2022.

[5] M. Rezayi and B. Nouri, "Personalization in mental health chatbots using deep learning," *IEEE Transactions on Computational Social Systems*, vol. 9, no. 2, pp. 234–245, 2022.

[6] A. S. Miner, A. Milstein, S. M. Schueller, R. Hegde, C. Mangurian, and E. Linos, "Smartphone-based conversational agents and responses to questions about mental health, interpersonal violence, and physical health," *JAMA Internal Medicine*, vol. 176, no. 5, pp. 619–625, 2016.

[7] Z. Xu, P. Zhang, and C. Wang, "Cognitive behavioral therapy (cbt) techniques in ai-powered mental health chatbots," *Journal of Digital Mental Health*, vol. 8, no. 2, pp. 78–95, 2020.

[8] E. Yom-Tov and A. V. Fisher, "Ai chatbots for crisis intervention: A systematic review," *Journal of Digital Psychology*, vol. 10, no. 1, pp. 45–62, 2021.

[9] R. K. Bailey and T. Patel, "Ethical considerations in ai-based mental health chatbots: Challenges and solutions," *Journal of AI Ethics*, vol. 4, no. 1, pp. 21–34, 2023.